🌟 What is Prompting?

Prompting is how you talk to AI models like ChatGPT to get the answers or outputs you want.

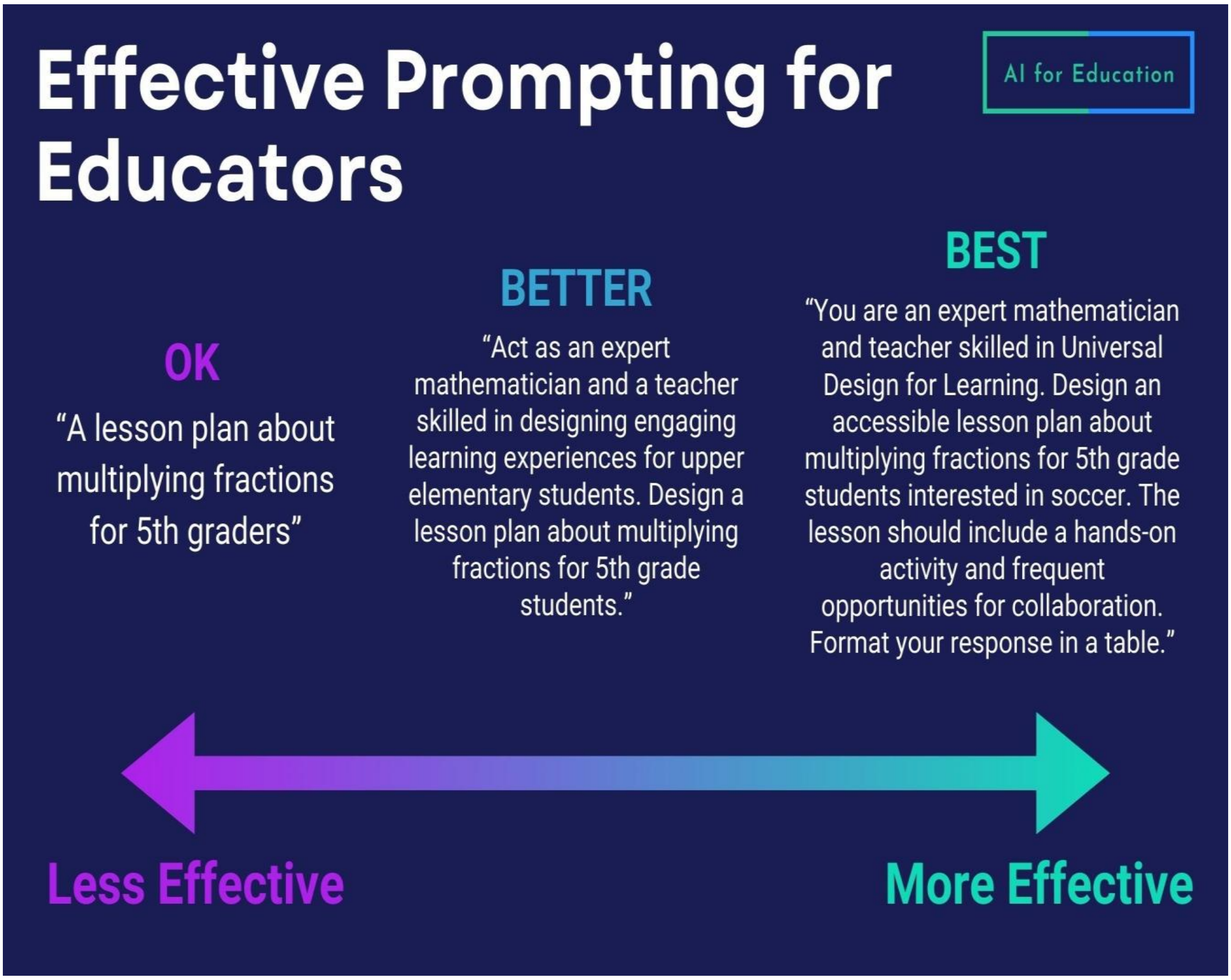
Think of a prompt as a question or instruction you give to an AI.

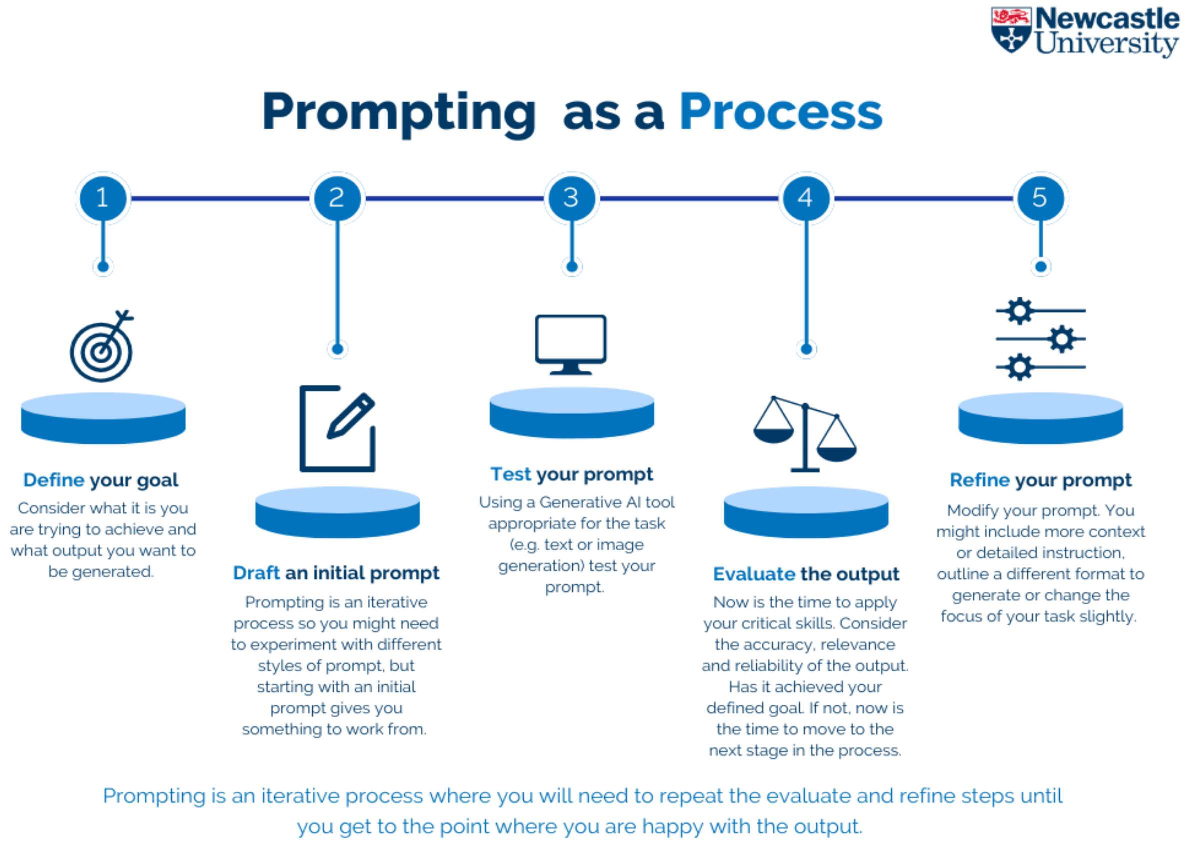
Example:

Prompt: "Write a short poem about the moon."

AI's Response:
"The moon glows bright in silent night,
A silver orb in pale moonlight..."

Just like giving commands to a computer or asking a question to a friend, you give prompts to an AI.





📌 Prompt Types

🔹 1. Zero-Shot Prompting

📘 **Definition**:
You give **no examples**, just a **direct instruction or question**.

🧠 The AI uses its **general knowledge** to respond.

✅ **Example:**

Prompt:
"What is the capital of France?"

AI Answer:
"Paris"

🔍 **When to Use**:

- The task is straightforward

- You want a quick answer
- You trust the AI to understand without help

🔹 2. Few-Shot Prompting

📘 **Definition**:
You give the AI **a few examples** of the task before asking your real question.

🧠 This helps the AI **learn the pattern** or **style** you're expecting.

✅ **Example:**

Prompt:
Translate these to French:

- Apple → Pomme
- Banana → Banane
- Orange → ?

AI Answer:
"Orange"

🔍 **When to Use**:

- Task has a format or pattern
- You want consistent answers
- The AI might not know what you expect

🔹 3. Chain-of-Thought Prompting

📘 **Definition**:
You ask the AI to **show its reasoning step-by-step** instead of jumping to the answer.

🧠 Helps the AI think like a human: "first this, then that, so the answer is…"

✅ **Example:**

Prompt:
"A shop sells pens at ₹15 each. If I buy 4 pens, how much do I pay? Show the steps."

AI Answer:

- One pen costs ₹15
- I buy 4 pens: 4 × ₹15 = ₹60
- So, I pay ₹60

🔍 **When to Use**:

- Math problems
- Logic puzzles
- Anything needing multiple steps

🔹 4. Few-Shot + Chain-of-Thought (Advanced Combo)

📘 **Definition**:
You give examples **and** ask the AI to **explain its reasoning**.

✅ **Example:**

Prompt:
Example 1:
Question: If 3 pencils cost ₹30, how much does 1 pencil cost?
Answer: ₹30 ÷ 3 = ₹10. So, 1 pencil costs ₹10.

Now you try:
Question: If 5 notebooks cost ₹100, how much does 1 notebook cost?

AI Answer:
₹100 ÷ 5 = ₹20. So, 1 notebook costs ₹20.

🔍 **When to Use**:

- Teaching AI complex tasks
- Very specific formatting

- Higher accuracy needed

Overview of Prompting Strategies

AI for Education

Strategy	Description	Use It For...	Example
Zero-Shot	Directly ask for an output without giving examples	Quick, general-purpose responses	“Write a paragraph describing the importance of renewable energy.”
Few-Shot	Providing examples of the desired output to guide the model’s response	Generating specific responses that need to conform to an established standard	“Write a course description for my Comp I class. Here is an example of a course description that I like:...”
Chain of Thought	Asking the model to break down its reasoning step-by-step before generating the final output	Encouraging a more thoughtful and accurate response	“Help me develop a process for onboarding new hires to my school district. Let’s think it through step-by-step...”
Explain-Then-Respond	Asking the model to explain a concept before applying it in context	Ensuring that foundational understanding is accurate	“Explain the key reasons that GenAI can prevent student learning. Then write a series of scenarios illustrating these concepts.”

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🎯 Prompt Tuning

🔧 **Prompt Tuning is a method used to improve the performance of a large language model (LLM) (like GPT) on specific tasks without changing the model itself.**

Instead of fine-tuning all the model's parameters (which is expensive and time-consuming), we tune a small prompt (like special keywords or vectors) to get better results for a specific task.

🧱 Let’s Build It Up Step-by-Step

✅ Step 1: Basic Prompting (Manual)

You write the prompt yourself:

"Summarize this article in 3 bullet points."

This works well, but:

- You have to guess what kind of prompt will work best
- It might not perform well on all inputs
- It's not optimized

✅ Step 2: Prompt Engineering (Manual Improvement)

You tweak the wording to improve results:

"Give me a brief, 3-bullet summary of the following article. Focus on key points."

This is better, but still trial-and-error. No training involved.

🔥 Step 3: Prompt Tuning (Automated Optimization)

Instead of writing and testing prompts manually, you let a machine learning process learn the best prompt to use for your task.

🧠 How it works:

1. You freeze the model (don't change it).
2. You create a special prompt, often made of learnable tokens (not real words).
3. You train just these tokens on your task (e.g., summarization, classification).
4. After training, the model performs better just by using the tuned prompt!

This is like giving the model a "magic phrase" that makes it perform better on your task.

📌 Example Comparison

Technique	What You Do	Changes the Model?	Training Needed?	Performance
Prompting	Ask a question directly	❌ No	❌ No	✅ Good
Prompt Engineering	Manually refine prompt wording	❌ No	❌ No	✅ Better
Prompt Tuning	Learn prompt using training data	❌ No (only prompt)	✅ Yes	✅ ✅ Very Good

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🔍 Real-World Use Cases

- Customer Support: Train prompts to better understand specific support issues
- Medical QA: Fine-tune prompts for more accurate medical reasoning
- Legal Document Analysis: Tailor prompts to extract legal clauses more precisely

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🧠 Advanced Concepts

1. Soft Prompts

- These are not actual words, but embedding vectors (numbers the model understands).
- Not human-readable.
- Trained like weights in a neural network.

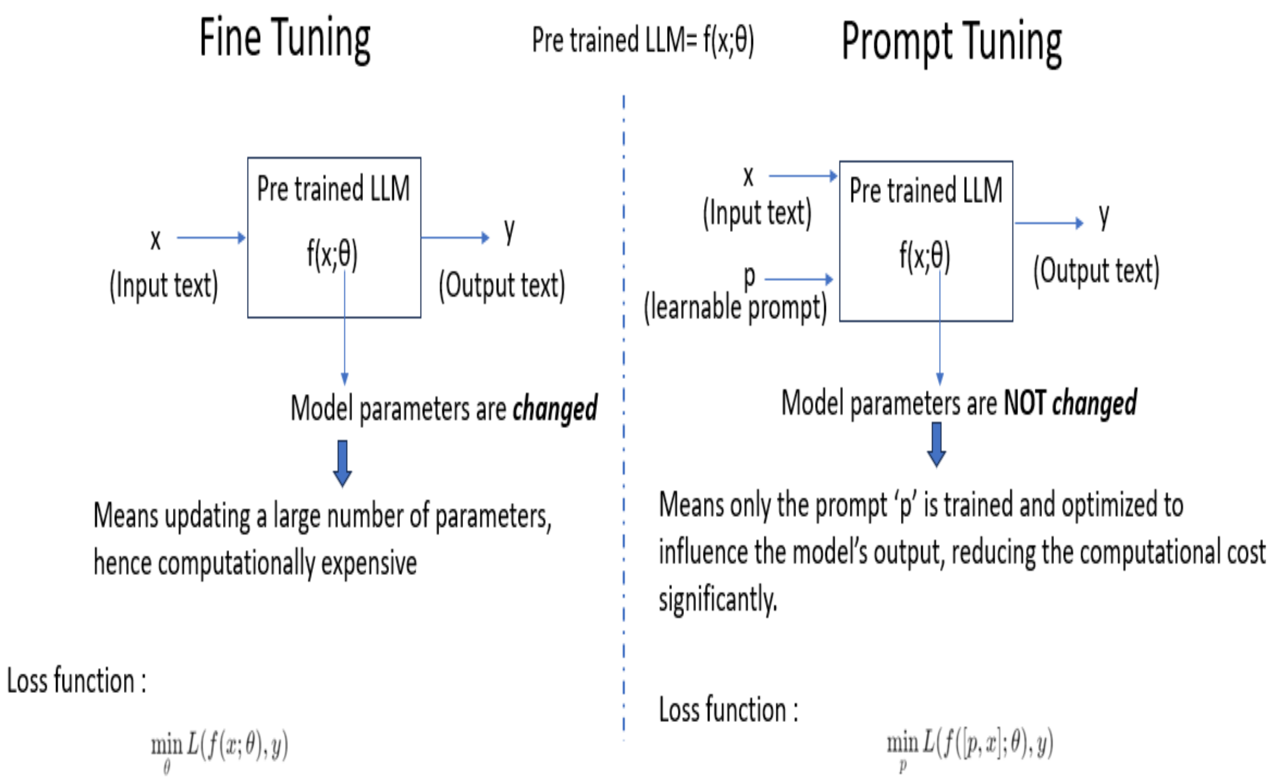
2. Prefix Tuning

- **A form of prompt tuning where a prefix (sequence of tokens) is trained and attached to every prompt.**
- **Often used with transformer models.**

3. P-Tuning / P-Tuning v2

- **Advanced techniques where prompts are inserted in multiple layers of the model (not just the input).**
- **Improve accuracy further with fewer parameters.**

⚖️ Prompt Tuning vs Fine-Tuning



Feature	**Prompt Tuning**	**Fine-Tuning**
Model weights	**Frozen (unchanged)**	**Changed (trained)**

Feature	Prompt Tuning	Fine-Tuning
Parameters updated	Few (only prompt vectors)	Millions (whole model)
Cost	Low	High
Flexibility	Narrow task focus	Can adapt more deeply
Speed	Fast to train	Slower

🧪 Tools & Frameworks Supporting Prompt Tuning

- OpenPrompt (open-source framework)
- HuggingFace PEFT (Parameter-Efficient Fine-Tuning)
- Google’s T5 / FLAN models
- PromptTune / P-Tuning by THUNLP